5.13.55

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Question: $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{pmatrix}$ and $\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $\mathbf{A}^{-1} = \frac{1}{6}(\mathbf{A}^2 + c\mathbf{A} + d\mathbf{I})$, then value of c and d are

1) $(-6, -11)$
2) $(6, 11)$
3) $(-6, 11)$
4) $(6, -11)$

Solution: Given the matrices

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{pmatrix} \quad \text{and} \quad I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{1}$$

and the relation $\mathbf{A}^{-1} = \frac{1}{6}(\mathbf{A}^2 + c\mathbf{A} + d\mathbf{I})$

First we Isolate the Unknowns
Starting with the given relation:

$$\mathbf{A}^{-1} = \frac{1}{6}\left(\mathbf{A}^2 + c\mathbf{A} + d\mathbf{I}\right) \tag{2}$$

Multiply both sides by 6 and rearrange to isolate the terms with the unknown scalars.

$$6\mathbf{A}^{-1} = \mathbf{A}^2 + c\mathbf{A} + d\mathbf{I} \tag{3}$$

$$c\mathbf{A} + d\mathbf{I} = 6\mathbf{A}^{-1} - \mathbf{A}^2 \tag{4}$$

We find the inverse using Gauss-Jordan elimination on the augmented matrix $\left(\mathbf{A} \mid \mathbf{I}\right)$.

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -2 & 4 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 \to R_3 + 2R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 6 & 0 & 2 & 1 \end{array}\right) \xrightarrow{R_2 \to R_2 - \frac{1}{6}R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2/3 & -1/6 \\ 0 & 0 & 6 & 0 & 2 & 1 \end{array}\right) \tag{5}$$

After scaling the last row, we get $\left(\mathbf{A} \mid \mathbf{I}\right)$. The resulting inverse matrix is:

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2/3 & -1/6 \\ 0 & 1/3 & 1/6 \end{pmatrix} \tag{6}$$

Now we Calculate the Square Matrix, $\mathbf{A}^2$ We compute $\mathbf{A}^2$ by matrix multiplication.

$$\mathbf{A}^2 = \mathbf{A}.\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{pmatrix} \tag{7}$$

Substituting the calculated matrices into Equation (2):

$$6\mathbf{A}^{-1} - \mathbf{A}^2 = 6\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2/3 & -1/6 \\ 0 & 1/3 & 1/6 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{pmatrix}$$
$$= \begin{pmatrix} 6 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 2 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{pmatrix}$$
$$= \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & -6 \\ 0 & 12 & -13 \end{pmatrix}$$

Now, set the left side of Equation (2) equal to this result:

$$c\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{pmatrix} + d\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & -6 \\ 0 & 12 & -13 \end{pmatrix} \tag{8}$$

$$\begin{pmatrix} c+d & 0 & 0 \\ 0 & c+d & c \\ 0 & -2c & 4c+d \end{pmatrix} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & -6 \\ 0 & 12 & -13 \end{pmatrix} \tag{9}$$

By equating corresponding elements, we can find the values of c and d.

From element (2, 3): $c = -6$

From element (2, 2): $c + d = 5 \implies -6 + d = 5 \implies d = 11$

The solution is therefore third option$(c, d) = (-6, 11)$.